

Markscheme

Mathematics: analysis and approaches
Practice question bank

70. (a)

The sum to infinity of a geometric sequence is 45, and the first term is 18.

Find the common ratio, and hence find the sum of the first 6 terms.

$$\frac{u_1}{1-r} = 45: \frac{18}{1-r} = 45 \quad (M1)$$

$$r = 0.6 \quad A1$$

$$S_6 = \frac{18(1-0.6^6)}{1-0.6} \quad (M1)$$

$$S_6 \approx 42.9 \quad A1$$

[6 marks]